



AIMS 11-01-004

Issue 4

Page 1 of 8

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AIMS
Airbus Material Specification

Honeycomb Cores, N-636 Polyamide Paper (meta and para-aramid),
Phenolic Resin Impregnated
Hexagonal
Material Specification

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Contents list

- 1 Scope
- 2 Application
- 3 Normative references
- 4 Technical requirements
- 5 Designation
- 6 Technical Specification

1 Scope

This Material Specification defines the requirements of honeycomb core made of N-636 polyamide paper (meta and para-aramid) impregnated with phenolic resin in hexagonal configuration in a cured state for aerospace application.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

Core material for the different types production of light-weight sandwich components.
Temperature range –55°C to 135 °C.

3 Normative references

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred shall be applied.

AIMS 11-01-000 Part 1	Airbus Material Specification; Honeycomb Cores, Phenolic Resin Impregnated - General Requirements. Technical Specification (TS)
AIMS 11-01-000 Part 2	Airbus Material Specification; Honeycomb Cores, Phenolic Resin Impregnated. Polyamide paper / Hexagonal / Over-expanded / Flexible. Qualification Programme - Release Testing
EN 2743	Aerospace series, Standard procedure for conditioning prior to testing
ABS5341	Airbus Standard; Honeycomb cores, N-636 Polyamide paper (meta and para-aramid), Phenolic resin impregnated. Hexagonal. - Semi finished Material Standard.

4 Technical requirements

4.1 General

Specification values are minimum individual except where stated

4.1.1 Material description

See table 1

Table 1: Material description

Characteristic	Requirements
Material type	honeycomb core, N636 polyamide paper (meta and para-aramid), phenolic resin impregnated
Configuration	hexagonal
State of material	cured
Material code (grades)	A to Y
Density	see table 2

4.1.2 Worklife and storage conditions

Worklife at max. 30 °C/75 % RH. shall be at least 12 months.

4.1.3 Processing conditions

The performance of all grades of honeycomb covered by this MS shall not be reduced by:

- expose to ≤ 248 °C for short periods (e.g. heat forming)
- autoclave or press curing processes with adhesives and/or prepregs at ≤ 185 °C.

4.1.4 Hazardous Substances

The relevant local health/safety and environmental regulations shall be fulfilled.

4.2 Specific

4.2.1 Physical-chemical properties

See table 2 and 3

4.2.2 Mechanical properties

See table 4

4.2.3 Burning behaviour

See table 5 (valid for all grades)

Table 2: Physical properties

(continued)

No.	Properties	Test method	Units	Requirements (minimum average values)					
Material code (grades) AIMS11-01-004				A	B	C	D	F	G
1	Cell size	TS Part 1 chapter: 5.3.1.5.2	mm	4.8	4.8	4.8	3.2	3.2	4.8
2	Paper thickness	TS Part 1 chapter: 5.3.1.5.3	µm	36	36	36	36	36	46
3	Density	TS Part 1 chapter: 5.3.1.5.5	kg/m ³	28	32	37	40	48	48
4a	Moisture absorption	TS Part 1 chapters: 5.3.1.5.6	%	< 5 ¹⁾					
4b				< 6 ²⁾					

(concluded.)

Table 2: Physical properties

No.	Properties	Test method	Units	Requirements (minimum average values)					
Material code (grades) AIMS11-01-004				L	P	T	W	U	Y
1	Cell size	TS Part 1 chapter: 5.3.1.5.2	mm	3.2	4.0	3.2	4.0	4.0	9.6
2	Paper thickness	TS Part 1 chapter: 5.3.1.5.3	µm	36	46	46	46	71	46
3	Density	TS Part 1 chapter: 5.3.1.5.5	kg/m ³	64	72	96	96	96	40
4a	Moisture Absorption ³⁾	TS Part 1 chapters: 5.3.1.5.6 a	%	<5 ¹⁾					
4b				<6 ²⁾					
1) Moisture absorption at 70 °C / 85 % R.H. 2) Moisture absorption 24 hours in water at R.T. (alternative method, see AIMS 11-01-000 Part 1). 3) Maximum individual values.									

Table 3: Physical properties: Paper properties (for information).

Properties	Test method	Units	Values based on qualification data		
Theoretical paper thickness	TS Part 1 chapter: 5.3.1.5.3	µm	36	46	71
Real paper thickness ¹⁾			41	48	71
Areal weight	ASTM D646	opsy	0.9 ± 0.07	1.20 ± 0.10	1.80 ± 0.15
		g/m ²	30.5 ± 2.4	40.7 ± 3.4	61.0 ± 5.1
1) Average values obtained in the qualification program process.					

Table 4: Mechanical properties

(continued)

No.	Properties	Test method	Test temp. [°C]	Units	Requirements							
Material codes (grades)					A	B	C	D	F	G		
1a	Compression strength	ASTM-C365 + TS Part 1 chapter: 5.3.1.6.1	23 ¹⁾	MPa	0.76	0.85	1.25	1.25	1.92	1.95		
1b			70 ¹⁾		0.65	0.76	1.10	1.10	1.73	1.76		
1c			135 ¹⁾		0.60	0.68	0.95	0.95	1.54	1.57		
1d			180 ¹⁾		0.53	0.60	0.80	0.75	1.34	1.37		
1e			70 ²⁾		0.60	0.68	0.95	0.95	1.54	1.57		
2a	L-direction	ASTM-C273 + TS Part 1 chapter: 5.3.1.6.2	23 ¹⁾		0.56	0.80	0.90	1.20	1.50	1.30		
2b			70 ¹⁾		0.50	0.72	0.81	1.05	1.35	1.18		
2c			135 ¹⁾		0.45	0.64	0.72	0.95	1.20	1.00		
2d			70 ²⁾		0.45	0.64	0.72	0.95	1.20	1.00		
2e			23 ¹⁾		0.40	0.48	0.50	0.70	0.80	0.65		
2f	70 ¹⁾		0.36		0.43	0.45	0.60	0.65	0.59			
2g	W-direction		135 ¹⁾		0.32	0.38	0.40	0.55	0.60	0.57		
2h	70 ²⁾		0.32		0.38	0.40	0.55	0.60	0.57			
2j	L-direction		23 ¹⁾		41.0	60.0	60.0	70.0	90.0	90.0		
2k			70 ¹⁾		37.0	54.0	55.0	65.0	81.0	81.0		
2l			135 ¹⁾		33.0	48.0	50.0	55.0	72.0	72.0		
2m			70 ²⁾		37.0	54.0	55.0	65.0	81.0	81.0		
2n			23 ¹⁾		21.0	24.0	30.0	36.0	45.0	50.0		
2o	Shear modulus		70 ¹⁾		18.0	22.0	27.0	33.0	41.0	43.0		
2p			W-direction		135 ¹⁾	17.0	19.2	24.0	30.0	36.0	40.0	
2q			70 ²⁾		18.0	22.0	27.0	33.0	41.0	43.0		
1) Dry, predrying for 1 h at 80 °C (according to EN 2743).												
2) Wet 70 °C/85 % R.H. according to AIMS 11-01-000 Part 2.												

(concluded)

Table 4: Mechanical properties

No.	Properties	Test method	Test temp. [°C]	Units	Requirements						
Material code (grades)					L	P	T	W	U	Y	
1a	Compression strength	ASTM-C365 + TS Part 1 chapter: 5.3.1.6.1	23 ¹⁾	MPa	3.30	4.20	6.50	6.50	6.50	1.50	
1b			70 ¹⁾		3.15	3.80	5.80	5.80	5.60	1.35	
1c			135 ¹⁾		2.80	3.40	5.10	5.10	4.90	1.20	
1d			180 ¹⁾		2.45	2.80	4.30	4.30	4.30	1.05	
1e			70 ²⁾		2.80	3.40	5.00	5.00	5.00	1.05	
2a	L-direction	ASTM-C273 + TS Part 1 chapter: 5.3.1.6.2	23 ¹⁾		2.00	1.90	2.70	2.70	3.20	1.00	
2b			70 ¹⁾		1.80	1.85	2.40	2.40	2.80	0.90	
2c			135 ¹⁾		1.60	1.60	2.20	2.20	2.15	0.80	
2d			70 ²⁾		1.60	1.60	2.40	2.40	2.60	0.80	
2e	Shear strength		23 ¹⁾		0.89	1.20	1.60	1.60	1.80	0.50	
2f			W-direction		70 ¹⁾	0.80	1.10	1.50	1.50	1.60	0.45
2g					135 ¹⁾	0.71	1.00	1.30	1.30	1.40	0.40
2h	70 ²⁾				0.71	1.00	1.50	1.50	1.70	0.40	
2j	L-direction		23 ¹⁾		114.0	120.0	150.0	150.0	170.0	22.0	
2k			70 ¹⁾		102.0	110.0	135.0	135.0	150.0	19.8	
2l			135 ¹⁾		91.0	95.0	125.0	125.0	135.0	17.6	
2m			70 ²⁾		102.0	110.0	130.0	130.0	140.0	17.6	
2n	Shear modulus		23 ¹⁾		60.0	65.0	75.0	75.0	85.0	16.0	
2o			W-direction		70 ¹⁾	54.0	60.0	68.0	68.0	75.0	14.4
2p					135 ¹⁾	48.0	55.0	60.0	60.0	70.0	12.8
2q	70 ²⁾				54.0	60.0	68.0	68.0	75.0	14.4	
1) Dry, pre drying for 1 h at 80 °C (according to EN 2743). 2) Wet 70 °C/85 % R.H. according to AIMS 11-01-000 Part 2.											

Table 5: Burning behaviour

No.	Properties			Test method	Unit	Requirements (max. average values)
Material code (grades)						AIMS11-01-004 (all)
1a	Flammability vertical 60s flaming	Burn length		TS Part 1 chapter: 5.3.1.7	mm	75
1b		After flame time			s	3
1c		After flame time of drips			s	0
2	Max. specific optical smoke density	F ¹⁾		TS Part 1 chapter: 5.3.1.7	-	50
3a	Max. concentration of smoke gas components	F ¹⁾	HCN	TS Part 1 chapter: 5.3.1.7	ppm	20
3b			CO			500
3c			NO/NO ₂			20
3d			SO ₂ /H ₂ S			20
3e			HF			0
3f			HCl			0
4a	Heat release rate max. within 5 min. ²⁾				kw/m ²	-
4b	Heat release at 2 min ²⁾				kw _{xmin} /m ²	-
1) F = Flaming Mode 2) Heat release testing shall be done within the qualification for low density honeycomb for interior parts only specimen and requirements shall be defined during qualification.						

5 Designation

See ABS5341

6 Technical specification

AIMS 11-01-000.

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 11/2001		New Standard- (Preliminary values)
2 06/2002		Extension of the of requirements (based on more qualification data)
3 10 / 2002		Added new core grade, revision of some requirements (Shear modulus / compression strength Grade L) (Shear modulus / compression strength Grade T) Added table 3
4 05 / 2005		Revision AIMS11-01-004 F, shear strength in w-direction